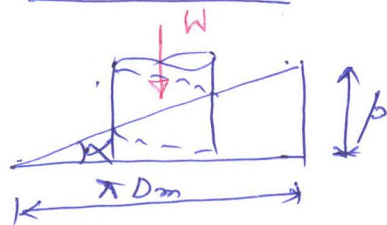
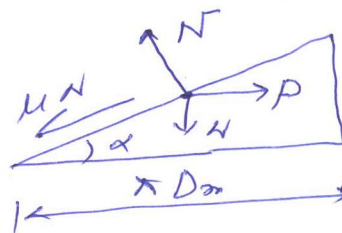


Screw Jack



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$$\sum F_x = 0: P = \mu N \cos \alpha + N \sin \alpha \quad \text{--- (1)}$$

$$\sum F_y = 0: N + \mu N \sin \alpha = N \cos \alpha.$$

$$\therefore N = N \cos \alpha - \mu N \sin \alpha \quad \text{--- (2)}$$

$$\textcircled{1} \div \textcircled{2} \quad \frac{P}{N} = \frac{\mu \cancel{N} \cos \alpha + \cancel{N} \sin \alpha}{\cancel{N} \cos \alpha - \mu \cancel{N} \sin \alpha} = \frac{\mu \cos \alpha + \sin \alpha}{\cos \alpha - \mu \sin \alpha}$$

$$\therefore \frac{P}{N} = \frac{\mu + \tan \alpha}{1 - \mu \tan \alpha} = \frac{\tan \phi + \tan \alpha}{1 - \tan \phi \tan \alpha} = \tan(\phi + \alpha)$$

$$\therefore \boxed{P = N \tan(\phi + \alpha)} \quad \text{--- (A)}$$

The effort P required to lower the load can be found

$$\text{as } \boxed{P = N \tan(\phi - \alpha)} \quad \text{--- (B)}$$

The moment required to raise the load N can be written

$$\text{as } \boxed{M = P \times \frac{D_m}{2} = \frac{N D_m}{2} \tan(\phi + \alpha) \text{ N-m}} \quad \text{--- (C)}$$

The moment required to lower the load N can be written

$$\text{as } \boxed{M = \frac{N D_m}{2} \tan(\phi - \alpha) \text{ N-m}} \quad \text{--- (d)}$$

When $\phi > \alpha$ eqⁿ (d) is (+)ive value

When $\phi < \alpha$ eqⁿ (d) is (-)ive value

When $\phi = \alpha$ eqⁿ (d) is zero value

for $\phi < \alpha$, (-)ive moment will act in the screw, hence the load N will descend down without any effort, which is the case of overhauling. So some amount of effort, hence moment will be required to stop the load to descend.

Hence $P = N \tan(\alpha - \phi)$ & $M = \frac{N D_m}{2} \tan(\alpha - \phi) \text{ N-m}$
when $\phi < \alpha$.

When $\phi = \alpha$

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Moment is zero. So the load will not descend down in the case $\phi = \alpha$.

When $\phi > \alpha$, (+)ive effort will be required to descend the load. This condition is known as SELF LOCKING CONDITION. This is desired for safe operation.

Efficiency of a square threaded screw used for screw Jack to raise the load.

$$\eta = \frac{\text{Output}}{\text{Input}} = \frac{W \cdot p}{P \cdot \pi D_m} = \frac{W \tan \alpha}{P} = \frac{W \tan \alpha}{W \tan(\phi + \alpha)} = \frac{\tan \alpha}{\tan(\phi + \alpha)}$$

$$\eta = \frac{\sin \alpha}{\cos \alpha} \times \frac{\cos(\phi + \alpha)}{\sin(\phi + \alpha)} = \frac{2 \sin \alpha \cos(\phi + \alpha)}{2 \cos \alpha \sin(\phi + \alpha)} = \frac{\sin(2\alpha + \phi) - \sin \phi}{\sin(2\alpha + \phi) + \sin \phi}$$

For η to be maximum $\sin(2\alpha + \phi)$ should be 1

$$\therefore \eta = \frac{1 - \sin \phi}{1 + \sin \phi} \quad \& \quad \sin(2\alpha + \phi) = 1 \quad \& \quad \alpha = 45^\circ - \frac{\phi}{2}$$

Efficiency of a square threaded screw used for screw Jack to raise the load.

As for safe operation (Self locking) $\phi > \alpha$. Hence, the η in this case can be found from when

$$\eta = \frac{\tan \alpha}{\tan(\phi + \alpha)} \quad \rightarrow \text{Limiting the value of } \alpha \text{ as } \phi$$

$$\eta \leq \frac{\tan \phi}{\tan(\phi + \phi)} = \frac{\tan \phi}{\tan 2\phi} = \frac{\tan \phi (1 - \tan^2 \phi)}{2 \tan \phi} \quad \rightarrow \text{for the case when } \alpha = \phi$$

$$\& \quad \eta \leq \left[\frac{1}{2} - \frac{\tan^2 \phi}{2} \right]. \text{ This relation shows that the}$$

η of the screw to raise the load can not exceed 50% for SELF LOCKING CONDITION.